

FoodLoop - AI Design & Workflow

1. Overview

Artificial Intelligence is a core component of FoodLoop, designed to simplify the process of listing surplus food while improving data quality and helping merchants make better selling decisions.

Rather than replacing user input, AI assists merchants by extracting product information from images, suggesting listing details, and recommending optimal discount prices based on product freshness and marketplace conditions.

The Project focuses on two primary AI capabilities:

- AI-assisted Product Recognition
- AI-assisted Dynamic Pricing

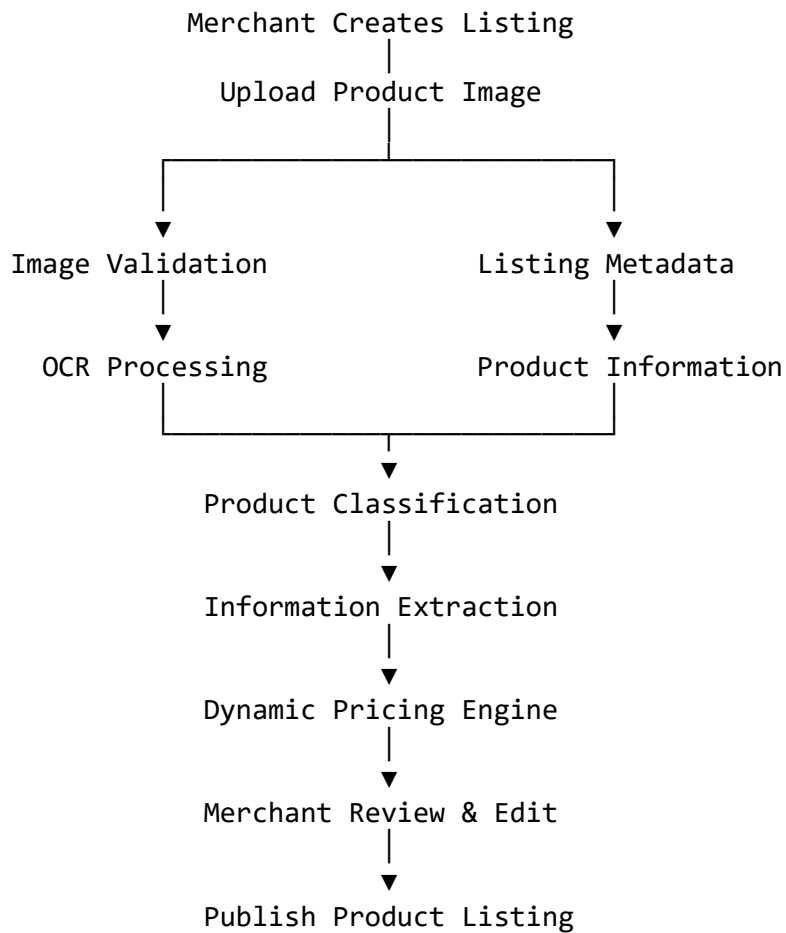
Both features are designed to reduce food waste while improving the merchant experience.

2. AI Objectives

The AI system aims to:

- Reduce the time required to create product listings.
- Minimize manual data entry.
- Improve listing consistency.
- Reduce human errors.
- Improve marketplace data quality.
- Recommend competitive pricing.
- Increase the likelihood of selling products before expiration.
- Assist platform moderation.

3. AI Architecture



4. AI Modules

4.1 Image Validation

Before any AI processing begins, uploaded images are validated.

Validation includes:

- Supported file type
- Image quality
- Minimum resolution
- File size
- Corrupted image detection

If validation fails, the merchant is prompted to upload a clearer image.

4.2 OCR (Optical Character Recognition)

OCR extracts readable text from product packaging.

Typical information includes:

- Product name
- Brand
- Ingredients
- Expiration date
- Production date

The extracted text is then used by later AI modules.

4.3 Product Classification

Using both the uploaded image and OCR results, the AI identifies the product.

Suggested outputs include:

- Product name
- Product category
- Food type
- Estimated brand

Example:

Input

Image of a yogurt container.

AI Suggestion

- Product: Greek Yogurt
 - Category: Dairy
 - Brand: Juhayna
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4.4 Information Extraction

Once the product has been identified, the AI prepares a draft listing.

Suggested information includes:

- Product title
- Category
- Expiration date

The merchant can modify any field before publishing.

4.5 Dynamic Pricing Engine

The Dynamic Pricing Engine recommends an appropriate selling price to maximize the chance of selling surplus food before it expires.

The system **does not automatically modify prices**. Instead, it provides pricing recommendations that merchants may accept or edit.

Pricing recommendations are generated using several factors, including:

- Original product price
- Product category
- Remaining shelf life
- Quantity available
- Historical sales performance
- Similar nearby listings
- Time of day
- Day of week
- Current marketplace activity

Outputs include:

- Suggested selling price
- Suggested discount percentage
- Estimated probability of sale
- Expected revenue

This helps merchants balance profitability with waste reduction.

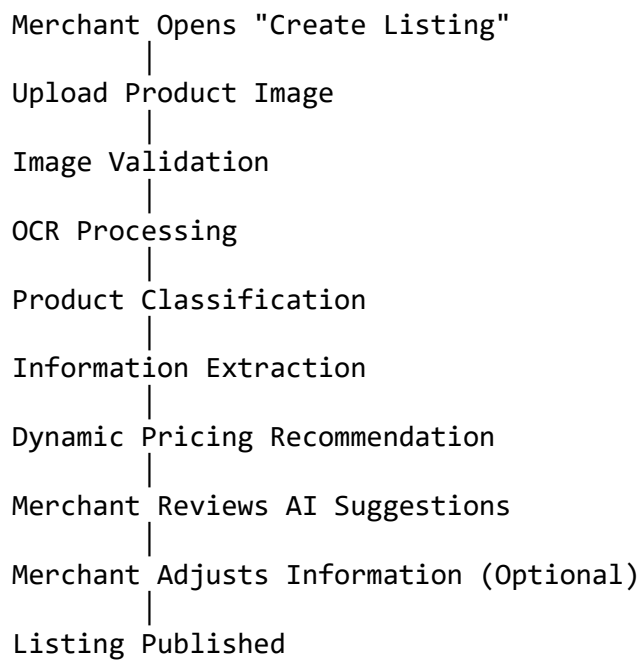
4.6 Confidence Scoring

Each AI prediction includes a confidence score.

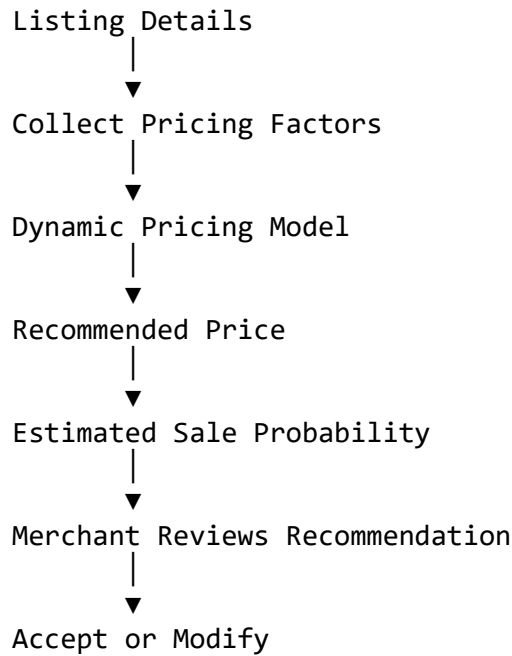
Confidence	System Behavior
≥ 90%	Automatically populate suggested fields
70–89%	Recommend values for merchant review
< 70%	Require manual input

Confidence scores ensure that uncertain predictions never become public listings without human review.

5. AI Listing Workflow



6. Dynamic Pricing Workflow



7. Human Validation

AI suggestions are intended to assist—not replace—the merchant.

Before publishing a listing, the merchant confirms:

- Product name
- Category
- Expiration date
- Description
- Price
- Quantity

This review process ensures that merchants remain responsible for the accuracy of published information.

8. AI Data Storage

The platform stores AI processing results for auditing and future model improvements.

Stored information includes:

- Uploaded image reference
- OCR output
- Product predictions
- Confidence score
- Pricing recommendation
- Merchant-approved values
- Processing timestamp

This data may also be used to evaluate AI performance over time.
